PHYS 338K Final Project Proposal: Op-Amp Based Function Generator

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Phys.338K

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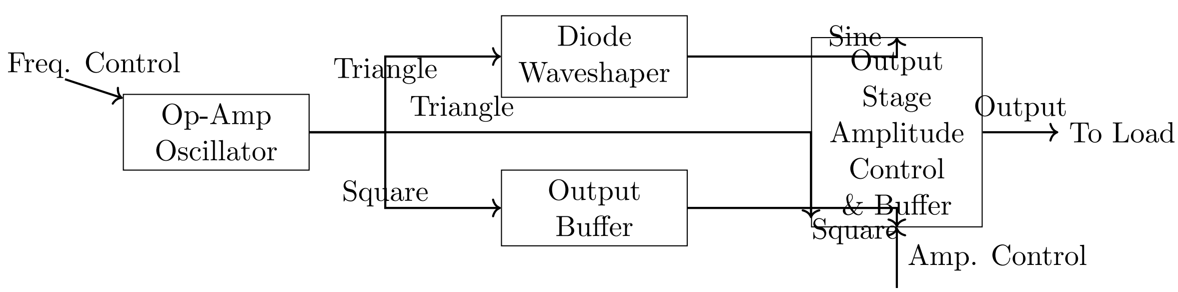
1. Project Description

This project will design and build a function generator capable of producing sine, square, and triangle waves. In response to feedback, the design will not use a dedicated waveform generator IC (e.g., ICL8038). Instead, it will be constructed from discrete op-amps to provide a more challenging and educational experience.

The core of the generator will be a two-op-amp oscillator that produces simultaneous square and triangle waves with amplitudes that remain constant when the frequency is tuned. This addresses a key limitation of simpler integrator-based designs. The triangle wave output will then be fed into a diode-based nonlinear waveshaping circuit to generate a low-distortion sine wave approximation. The final outputs will be buffered to provide a low-impedance source.

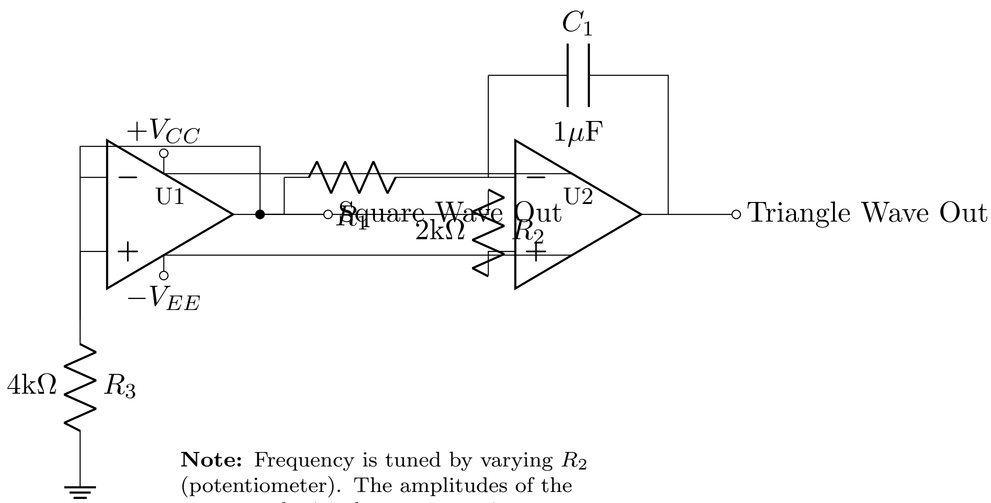
2. Circuit Diagram

Block Diagram:



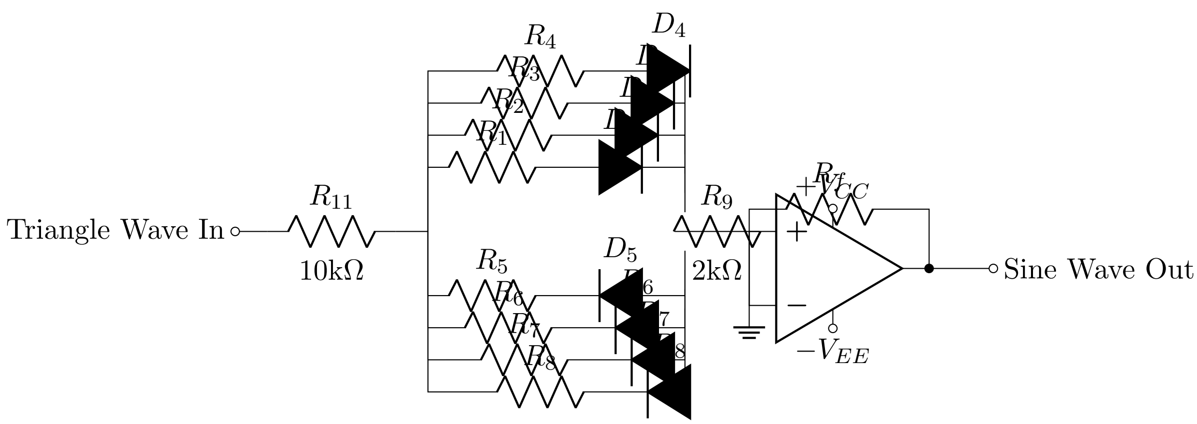
Detailed Schematic:

Part (a): Two Op-Amp Triangle/Square Wave Oscillator (Constant Amplitude)



Note: Frequency is tuned by varying R_2 (potentiometer). The amplitudes of the square and triangle waves remain constant.

Part (b): Diode-Based Triangle-to-Sine Wave Shaper



Circuit (b) Description: This network uses biased diodes (D_1-D_8) to progressively clip the peaks of the triangle wave, rounding it into a sine wave. Resistors R_1-R_8 set the bias points for the diodes. The op-amp sums the currents from the network.

3. Revised Parts List

- **Op-Amps:** 3-4 x LM741, TL081, or similar general-purpose op-amps
- **Diodes:** 8 x 1N4148 or 1N914 small-signal switching diodes
- **Resistors:** Multiple values for oscillator, waveshaper, and biasing (e.g., 1kΩ, 2kΩ, 4kΩ, 6.8kΩ, 10kΩ)
- **Capacitors:** 1μF timing capacitor and decoupling capacitors
- **Potentiometers:** One for frequency control (R_2), one for output amplitude control
- **Switch:** Single Pole, 3-Position (SP3T) rotary switch for waveform selection

- **Power Supply:** ±9V or ±12V dual rail supply
- **Output Connector:** BNC or 3.5mm audio jack
- **Breadboard/PCB:** For prototyping and final assembly